As an example, the more precisely you know where a particle is, the less precisely you can possibly know its speed. Similarly, the more precisely you know how fast a particle is moving, the less you can possibly know about where it is. Quantum theory thereby sets up its own duality: you can determine with precision certain physical features of the microscopic realm, but in so doing you eliminate the possibility of precisely determining certain other, complementary features.

To understand why, let's follow a rough description developed by Heisenberg himself, which, while incomplete in particular ways that we will discuss, does give a useful intuitive picture. When we measure the position of any object, we generally interact with it in some manner. If we search for the light switch in a dark room, we know we have located it when we touch it. If a bat is searching for a field mouse, it bounces sonar off its target and interprets the reflected wave. The most common instance of all is locating something by seeing it—by receiving light that has reflected off the object and entered our eyes. The key point is that these interactions not only affect us but also affect the object whose position is being determined. Even light, when bouncing off an object, gives it a tiny push. Now, for day-to-day objects such as the book in your hand or a clock on the wall, the wispy little push of bouncing light has no noticeable effect. But when it strikes a tiny particle like an electron it can have a big effect: as the light bounces off the electron, it changes the electron's speed, much as your own speed is affected by a strong, gusty wind that whips around a street corner. In fact, the more precisely you want to identify the electron's position, the more sharply defined and energetic the light beam must be, yielding an even larger effect on the electron's motion.

This means that if you measure an electron's position with high accuracy, you necessarily contaminate your own experiment: the act of precision position measurement disrupts the electron's velocity. You can therefore know precisely where the electron is, but you cannot also know precisely how fast, at that moment, it was moving. Conversely, you can measure precisely how fast an electron is moving, but in so doing you will contaminate your ability to determine with precision its position. Nature has a built-in limit on the precision with which such complementary features can be determined. And although we are